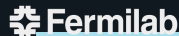

DUNE detector-wide interlocks

A draft framework for protection, operating exclusions, and data quality

Manuel Arroyave

DUNE detector-coordination discussion draft – 4 September 2026



DUNE needs one rulebook for cross-detector constraints

THE PROPOSAL

Keep one controlled interaction register.

Each row names the condition, consequence, response, owner, recovery, and test. Every public matrix is a view of that same row.

PROTECTION

Stop credible damage

Trip or inhibit the hazardous energy path.

OPERATIONS

Keep incompatible modes apart

Deny the second request or require a named hold.

DATA QUALITY

Keep useful data honest

Allow the activity and publish the affected interval.

Not every “not allowed” condition is a hardware interlock

THE CENTRAL DISTINCTION

The response must follow the consequence.

“Incompatible” has historically covered three very different situations.

CREDIBLE DAMAGE

Unsafe cryogenic condition with drift HV

Use an independent trip or inhibit.

OPERATING EXCLUSION

Camera or external laser with PDS

Deny the second request; do not invent a PDS trip.

DATA QUALITY

Purity monitor during PDS acquisition

Allow the overlap and tag the data.

Every rule answers two questions

1 · WHAT IS AT STAKE?

Consequence class

PS / ED Personnel, facility, or equipment protection

OP Operating conflict or loss of availability

DQ Physics or data-quality contamination

OK / UNK Compatible, or not yet classified

2 · HOW IS IT HANDLED?

Enforcement class

FSS / AUTO Independent safe state, trip, or inhibit

SW / HOLD Software denial or named authorization

TAG / NONE Allow, record, or take no special action

TBD Not creditable until owners resolve it

TAKEAWAY A cell such as ED/AUTO [P-002] states both the reason and the response.

Only a credible damage case can demand an automatic trip

THE TEST

Describe the damage before prescribing the trip.

A red cell is evidence of concern; it is not yet a protection requirement.

01 Protected asset

Name exactly what can be damaged.

02 Damage mechanism

State the physical initiating condition.

03 Safe action

Name scope, timing, latch, and reset.

04 Independent proof

Map the final element and fault-test it.

The protection matrix names the trigger and safe action

Initiating condition	Required safe action	Return condition	Rule
Cryogenic HV permit is false, low, stale, or abnormal while drift HV is on	Block enable or trip the affected drift HV; latch	Cause clear, inspection, fresh permit, deliberate ramp	P-001-003
Cathode overcurrent, streamer, or drift-HV shut-down	Trip cathode HV and invoke the mapped field-cage action	Inspect, reconcile states, restore in approved order	P-004/005
One bias channel exceeds its approved electrical or thermal envelope	Remove the smallest approved channel or group	Cause clear, authorized reset, fresh readback	P-018
Fire or damaging rack condition	Apply the approved facility or rack cause-and-effect action	Facility release and protection proof test	P-006-009

Source: [SC-HVS interface](#), [EDMS 3315615/1](#)

Camera and external-laser requests share one PDS mutex

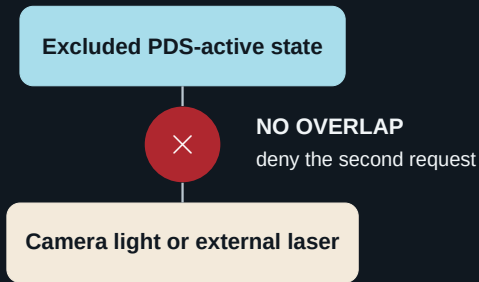


Diagram. The mutex works in both command orders. Laser personnel safety remains an independent facility-safety function.

0-001, 0-002: operating mutex. S-002: laser personnel safety.

01 Reject the new request

The activity already in control keeps the optical region.

02 Verify the return state

Confirm light, emission, and shutter off before PDS return.

03 Record accidental overlap

Invalidate or degrade the affected interval and alert operations.

Purity-monitor activity stays in the data-quality lane

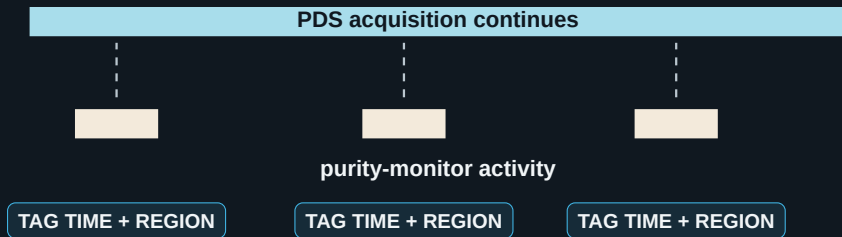


Diagram. The overlap affects measurement quality; it does not require PDS bias to be removed.

DATA-QUALITY RULE Q-001

Allow the overlap and publish the quality flag

Keep the data. Schedule a quiet window only for measurements that need clean PDS data.

The PDS matrix uses three different responses

Interaction with PDS	Coexist?	Admission or control	Data and return
Camera illumination / external LEDs	NO	Deny the second request	Tag overlap; verify light off
Ionisation or alignment laser	NO	Deny second request; separate laser FSS	Tag overlap; verify emission and shutter off
Purity monitor	YES	Allow; quiet windows optional	Tag affected time and region; no bias action
PDS-owned calibration light	Calibration mode	Versioned PDS/DAQ sequence	Tag pulse, configuration, and region
Power-over-Fiber for applicable VD PDS	REQUIRED	Exempt from external-laser mutex	Keep its own electrical and thermal protection

0-001, 0-002, Q-001, Q-002, C-001. Exact "PDS active" state remains an owner decision.

HV and bias states need explicit, scoped actions

Event	Cathode / field cage	APA or CRP bias	PDS / DAQ
Cryogenic HV-ready is false before enable	Block cathode-HV ramp	Preserve actual states; use approved recipe	Not ready; publish cause
Low LAr, relief, or approved boiling risk	Trip and latch cathode HV; mapped FC action	Owner-approved action only	Stop or tag; notify
Cathode overcurrent or streamer	Trip cathode HV; mapped FC action	Coupled action remains open	Inhibit or tag; save waveform
One bias channel exceeds its safe envelope	No blanket cathode action	Remove smallest approved channel or group	Not ready; publish cause
Fire condition	Apply approved HVS cause and effect	Apply approved HVS cause and effect	Stop; facility record

TAKEAWAY One event does not justify an unreviewed detector-wide trip.

A cathode trip does not yet imply a detector-wide trip

WHAT IS CONTROLLED

Cathode shutdown invokes the mapped field-cage action

The current HVS interface states this coupling. It also requires notification to detector operations.

WHAT REMAINS OPEN

The coupled action for the rest of the detector

No available controlled requirement establishes one universal response for APA/CRP bias, readout electronics, or PDS/PoF.

OPEN RULE P-019

Resolve HD and VD separately

Test both orders; measure transients; approve scope, timing, and recovery.

Source: [SC-HVS interface](#), [EDMS 3315615/1](#)

ProtoDUNE already used human interlocks

PDS + CAMERA LIGHT

People waited for lights to be off

NP02 PDS work was coordinated around camera and LED state.

PURITY MONITOR + DAQ

A state once inhibited triggers

A missing purity-monitor status also left the historical inhibit unreleased.

PDS + LASER

People handed the detector back

NP04 returned PDS only after laser emission or shutter state was clear.

HV + DETECTOR STATES

People followed explicit sequences

Shifters coordinated cathode, bias, PDS, electronics, and DAQ transitions.

WHAT THIS EVIDENCE MEANS

The workflow is real; the protection claim is not yet proven

History identifies candidate rules. Analysis and testing certify protection.

Human practice becomes a requirement, not a copy of the chat

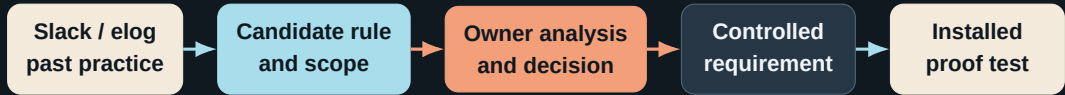


Diagram. Historical behavior starts the conversation. A rule becomes creditable only after ownership, controlled requirements, and installed verification.

TAKEAWAY No unowned Slack instruction becomes detector logic.

Routine coordination needs machine-readable state

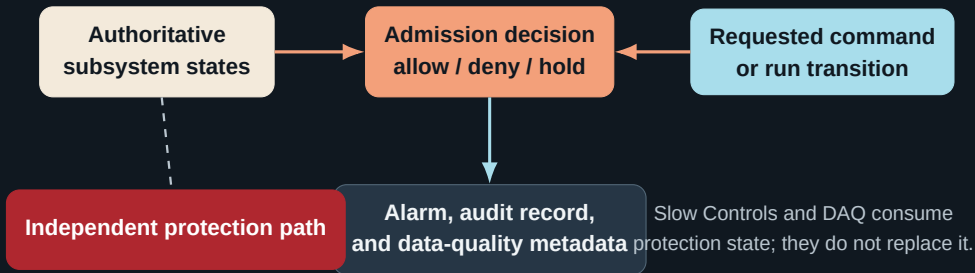


Diagram. Operational admission can be software. Credited equipment or personnel protection remains independent of ordinary detector software.

Test both command orders and every stale-state case

Test	Expected decision	Evidence to keep
State A active; request B	Deny, hold, or allow exactly as the rule says	Request, authoritative state, decision, timestamp
State B active; request A	Same rule works in the reverse order	Reverse-order decision and return state
Simultaneous requests	Deterministic priority; no race to an unsafe state	Ordered event log and final states
Missing, stale, or contradictory state	Fail to the approved conservative decision	Quality, timeout, alarm, and owner response
Restart during an active condition	Reconcile actual state before accepting commands	Startup snapshot and explicit recovery

The lifecycle changes enforcement—not the rule

INSTALLATION

Isolation comes first

Default off. Use work control, lockout/tagout, and independent verification. Record the installed mapping before any protection claim.

COMMISSIONING + INTEGRATION

Bound every exception

Name the test owner, scope, recipe, expiry, and recovery. Human interlocks remain visible while interfaces mature.

RUN

Automate routine decisions

Use autonomous protection for damage, software admission for exclusions, and metadata for data-quality effects.

TAKEAWAY The rule ID and physical consequence remain stable across all three stages.

One register produces several audience views

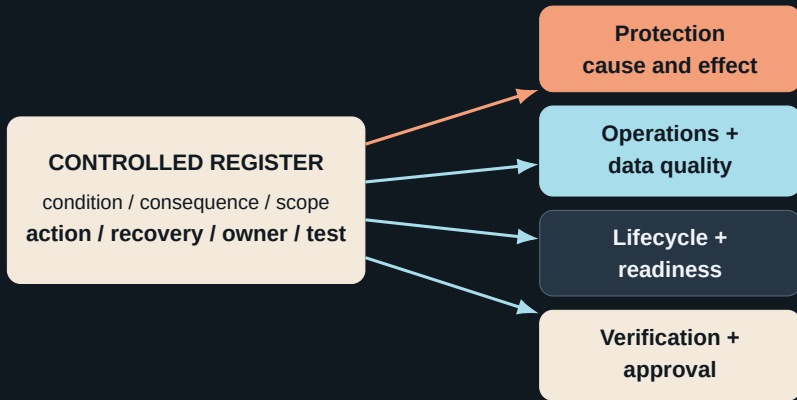


Diagram. Public matrices stay readable because detailed requirements live once in the controlled register. Every visible cell carries a stable rule ID.

Evidence has a clear authority ladder

Evidence source	What it establishes	What it cannot establish alone
Slack, elog, and shift instructions	Past behavior Who waited, what stayed off, and how handoff worked.	Automatic protection A human habit is not a credited trip.
Approved ICD or requirement	Required interface behavior Named states, notifications, and candidate actions.	The complete hazard case Boundaries and response budgets may remain open.
Hazard and electrical analysis	Why protection is needed Damage mechanism, safe state, scope, and timing.	Installed performance The real path still needs testing.
Installed proof or fault test	What the detector really does Ordering, latency, readback, and recovery.	Policy ownership Owners must still accept the result.

Approve four things before implementation

01 Approve the two-axis grammar

Every row states consequence and enforcement separately.

02 Approve the first operating rules

PDS/light mutex; purity-monitor data tag; PoF exception.

03 Close the open coupling

Resolve P-019 separately for HD and VD.

04 Name owners and proof tests

Set thresholds, scope, timing, reset, and the pass result.

WHAT “DONE” LOOKS LIKE

Every rule has a reason, an owner, and a tested recovery

Until then, this remains a discussion draft—not detector logic.